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PARAMETER OR NON-PARAMETER?

These Models assume the relationship between variables.

These Models are often simpler and faster to learn.

The Models require a large amount of data for accurate predictions.

These Models are computationally efficient when dealing with large datasets.

These Models' lack of flexibility might lead to underfitting if the data has a more complex structure.

Parametric

Non-Parametric

K

N

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PARAMETRIC OR NON-PARAMETRIC?

Parametric

These Models are often simpler and faster to learn.

These Models assume a specific form for the relationship between variables.

These Models are more computationally efficient when dealing with large datasets.

These Models' lack of flexibility might lead to underfitting if the data has a more complex structure.

Non-Parametric

These Models require a larger amount of data for accurate predictions.

KNN

BIAS & VARIANCE [FLASHBACK TO L#1]

Bias— Bias refers to the error from simplifying a real-world problem with an overly simple model. High bias can cause the model to overlook important relationships between input features and output predictions.

With high bias, adding more data might not help as the model is too basic to grasp data complexities.

Variance— Reflects the model's responsiveness to small fluctuations or noise in training data. A model with high variance excels on its training data by accounting for random noise, but falters on new, unseen data (overfitting).

With high variance, adding more data can often boost performance since it helps "average out" or reduce the noise captured by the model.

